

SHYAM SUNDAR DOMAKONDA

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PROFESSIONAL SUMMARY:

Qualified Data Scientist/Data Analyst with over 5+ years of experience in Data Science and Analytics including Artificial Intelligence / Deep Learning / Machine Learning, Data Mining and Statistical Analysis.

- Strong skills in statistical methodologies such as A/B test, experiment design, hypothesis test, ANOVA.
- Extensively worked on Python (Numpy, Pandas, Matplotlib, NLTK and Scikit-learn)/
- Solid ability to write and optimize diverse SQL queries, working knowledge of RDBMS like SQL Server 2008, NoSql databases like MongoDB 3.2.
- Involved in the entire data science project life cycle and actively involved in all the phases including data extraction, data cleaning, statistical modeling and data visualization with large data sets of structured and unstructured data, created ER diagrams and schema.
- Experienced with machine learning algorithm such as logistic regression, random forest, XG boost, KNN, SVM, neural network, linear regression, lasso regression and k – means.
- Implemented Bagging and Boosting to enhance the model performance.
- Developed API libraries and coded business logic using C#, XML and designed web pages using .NET framework, C#, Python, Django, HTML, AJAX.
- Strong experience in Big Data technologies like Spark 1.6, Spark sql, pySpark, Hadoop 2.X, HDFS, Hive 1.X.
- Experience in visualization tools like, Tableau 9.X, 10.X for creating dashboards.
- Excellent understanding Agile and Scrum development methodology.
- Used the version control tools like Git 2.X and build tools like Apache Maven/Ant.
- Passionate about gleaning insightful information from massive data assets and developing a culture of sound, data-driven decision making.
- Ability to maintain a fun, casual, professional and productive team atmosphere.
- Experienced the full software life cycle in SDLC, Agile, Devops and Scrum methodologies including creating requirements, test plans.
- Skilled in Advanced Regression Modeling, Correlation, Multivariate Analysis, Model Building, Business Intelligence tools and application of Statistical Concepts.
- Proficient in Predictive Modeling, Data Mining Methods, Factor Analysis, ANOVA, Hypothetical testing, normal distribution and other advanced statistical and econometric techniques.
- Developed predictive models using Decision Tree, Random Forest, Naïve Bayes, Logistic Regression, Social Network Analysis, Cluster Analysis, and Neural Networks.
- Experienced in Machine Learning and Statistical Analysis with Python Scikit-Learn.
- Experienced in Python to manipulate data for data loading and extraction and worked with python libraries like Matplotlib, Numpy, Scipy and Pandas for data analysis.
- Worked with complex applications such as R, SAS, Matlab and SPSS to develop neural network, cluster analysis.
- Strong C#, SQL programming skills, with experience in working with functions, packages and triggers.
- Expertise in transforming business requirements into analytical models, designing algorithms, building models, developing data mining and reporting solutions that scales across massive volume of structured and unstructured data.
- Skilled in performing data parsing, data ingestion, data manipulation, data architecture, data modelling and data preparation with methods including describe data contents, compute descriptive statistics of data, regex, split and combine, Remap, merge, subset, reindex, melt and reshape.
- Experienced in Visual Basic for Applications and VB programming languages C#, .NET framework to work with developing applications.
- Worked with NoSQL Database including Hbase, Cassandra and MongoDB.
- Experienced in Big Data with Hadoop, HDFS, MapReduce, and Spark.
- Experienced in Data Integration Validation and Data Quality controls for ETL process and Data Warehousing using MS Visual Studio SSIS, SSAS, SSRS.
- Proficient in Tableau and R-Shiny data visualization tools to analyze and obtain insights into large datasets, create visually powerful and actionable interactive reports and dashboards.
- Automated recurring reports using SQL and Python and visualized them on BI platform like Tableau.
- Worked in development environment like Git and VM.
- Excellent communication skills. Successfully working in fast-paced multitasking environment both independently and in collaborative team, a self-motivated enthusiastic learner.

TECHNICAL SKILLS:

Category	Skills/Tools
Programming Languages	Python: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch. R: ggplot2, dplyr, caret. Java: Building scalable AI systems, enterprise apps. C++: TensorFlow, PyTorch performance optimization. Julia: High-performance numerical computing.
Machine Learning Algorithms	Supervised Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forest, XGBoost, LightGBM. Unsupervised Learning: K-Means, DBSCAN, Hierarchical Clustering, PCA, t-SNE, UMAP. Reinforcement Learning: Q-Learning, DQN, Policy Gradient Methods. Neural Networks: FNN, CNN, RNN, LSTMs, Transformers (BERT, GPT). Ensemble Methods: Random Forest, Gradient Boosting (XGBoost, LightGBM), Stacking.
Deep Learning Frameworks	TensorFlow: Model building, training, deployment. Keras: High-level neural network API. PyTorch: Dynamic computation graphs, research focused. MXNet: Scalable deep learning. Hugging Face: NLP. LLM's & Transformer: GPT 3.5, ChatGPT, Llama, Mistral AI, BERT, RoBERTa. Discrete Mathematics: Graph theory, decision trees, combinatorics.
Deployment & Scalability	Model Deployment: Docker, Flask, FastAPI, TensorFlow Serving. Cloud Platforms: AWS (SageMaker, EC2), Google Cloud (Vertex AI), Azure ML. Scalability: Apache Spark, Dask, Ray, distributed computing.
Big Data Technologies	Apache Hadoop: Distributed storage and processing. Apache Spark: Large-scale data processing and model training. NoSQL Databases: MongoDB, Cassandra for unstructured data. Relational Databases: MySQL, PostgreSQL. Graph Database: Neo4j. Version Control, Containerization & Orchestration: Git, GitHub, Docker, Kubernetes, Airflow.

PROFESSIONAL EXPERIENCE:

Client: Walmart Global Tech

Role: AI/ML Data Scientist

Responsibilities:

Bentonville, AR, USA

January 2024 - Present

- Performed Data Profiling to learn about behavior with various features of USMLE examinations of various student patterns.
- Led end-to-end data science projects including problem formulation, data preprocessing, model development, and deployment, resulting in [specific measurable outcomes, e.g., a 15% increase in sales, reduced churn by 10%].
- Built and deployed machine learning models using Python (Scikit-learn, TensorFlow, PyTorch) to predict customer behavior, driving business strategies and product improvements.
- Conducted in-depth exploratory data analysis (EDA) on large datasets (SQL, Pandas), uncovering valuable insights for marketing and product teams.
- Designed and implemented time series models (ARIMA, Prophet) to forecast sales trends, optimizing inventory and supply chain management.
- Evaluated models using Cross Validation, Log loss function, ROC curves and used AUC for feature selection and elastic technologies like Elastic Search, Kibana etc.
- Implemented statistical modeling with XG Boost machine learning software package using Python to determine the predicted probabilities of each model.
- Created master data for modelling by combining various tables and derived fields from client data and students LORs, essays and various performance metrics.
- Formulated a basis for variable selection and Grid Search, KFold for optimal hyper parameters
- Utilized Boosting algorithms to build a model for predictive analysis of student’s behaviour who took USMLE exam apply for residency.
- Designed and deployed scalable ML pipelines using Dataproc for data preprocessing, BigQuery for feature engineering, and Vertex AI for model training, reducing model development time by 40%.

- Built a centralized Data Lake in GCS, integrating structured and unstructured data with BigQuery, enabling faster analytics and reducing query execution time by 30%.
- Fine-tuned and deployed LLMs on Vertex AI, leveraging GCS for training data storage and BigQuery for real-time feature extraction, improving response accuracy by 25%.
- Used numpy, scipy, pandas, NLTK (Natural Language Processing Toolkit), matplotlib to build the model.
- Formulated several graphs to show the performance of the students by demographics and their mean score in different USMLE exams.
- Developed and optimized XGBoost and Random Forest models for predictive analytics, improving customer segmentation accuracy by 15%.
- Engineered advanced feature selection techniques, reducing model complexity and increasing XGBoost efficiency while maintaining high precision.
- Tuned hyperparameters for Random Forest using GridSearchCV and Bayesian Optimization, enhancing model performance and reducing overfitting.
- Created Bash/PowerShell scripts to deploy ML models on Azure ML, Vertex AI, and SageMaker, automating model updates and versioning, improving deployment efficiency by 40%.
- Designed an MLOps pipeline using SageMaker Pipelines, Lambda, and S3, automating model retraining based on data drift, ensuring continuous model accuracy above 90%.
- Trained and optimized ML models in Databricks MLflow, implementing hyperparameter tuning (Grid Search, Bayesian Optimization) and distributed model training, reducing training time by 40%.
- Integrated Databricks with GCP/AWS for end-to-end ML pipelines, deploying models via Databricks AutoML and monitoring performance using MLflow, ensuring seamless production-ready deployment.
- Designed and optimized ETL pipelines using AWS Lambda to automate data ingestion from multiple sources into AWS S3 and transformed raw data for analytics in Amazon Redshift, improving data processing speed by 30%.
- Fine-tuned GPT-4, LLaMA 2, and Falcon on domain-specific data using LoRA (Low-Rank Adaptation) and PEFT (Parameter Efficient Fine-Tuning) techniques, improving model accuracy by 35% while reducing compute costs.
- Built and deployed custom chatbots and text-generation models using LLMs like T5 and BLOOM via Hugging Face Transformers and LangChain, enhancing customer support automation by 40%.
- Benchmarked LLM performance using BLEU, ROUGE, and perplexity scores, optimizing response quality for a retrieval-augmented generation (RAG) system, improving relevance by 20%.
- Designed data pipelines for pretraining and fine-tuning large-scale LLMs by cleaning and processing 100+ TB of text and code data using PySpark, AWS S3, and Databricks, reducing training time by 30%.
- Developed multi-modal AI models integrating text and image generation using GPT-4V (vision), CLIP, and Stable Diffusion, enabling advanced content creation workflows for marketing automation.
- Developed and deployed custom generative AI models for text, image, and code generation using GPT-4, Stable Diffusion, and DALL·E, enhancing content automation by 40%.
- Designed and implemented RAG pipelines using LangChain, FAISS, and Vector Databases (Weaviate, Pinecone, ChromaDB) to improve contextual accuracy in AI-driven applications.
- Fine-tuned LLMs like LLaMA, Falcon, and Mistral on proprietary datasets using LoRA, QLoRA, and PEFT, reducing computational cost by 35% while improving domain-specific performance.
- Deployed fine-tuned GenAI models on AWS SageMaker, Vertex AI, and Azure OpenAI, optimizing inference speed by 50% through model quantization and distillation.
- Application of various Artificial Intelligence (AI)/machine learning algorithms and statistical modeling like decision trees, text analytics, natural language processing (NLP), supervised and unsupervised, regression models.
- Used Principal Component Analysis in feature engineering to analyze high dimensional data.
- Performed Data Cleaning, features scaling, features engineering using pandas and numpy packages in python and build models using deep learning frameworks.
- Created deep learning models using Tensorflow, Pytorch and keras by combining all tests as a single normalized score and predict residency attainment of students.
- Used XGBoost classifier if the feature is a categorical variable and XGBoost regress or for continuous variables and combined it using Feature Union and Function Transformer methods of Natural Language Processing.
- Used One vs Rest Classifier to fit each classifier against all other classifiers and used it on multiclass classification problems.
- Implemented application of various machine learning algorithms and statistical modeling like Decision Tree, Text Analytics, Sentiment Analysis, Naive Bayes, Logistic Regression and Linear Regression using Python to determine the accuracy rate of each model.
- Created and designed reports that will use gathered metrics to infer and draw logical conclusions of past and future behavior.
- Generated various models by using different machine learning and deep learning frameworks and tuned the best performance model using Signal Hub.
- Created data layers as signals to Signal Hub to predict new unseen data with performance not less than the static model build using deep learning framework like tensorflow and Pytorch.
- Designed Docker-based environments for data preprocessing, model training, and inference, standardizing dependencies and

eliminating environment-related failures.

Client: University of North Texas
Role: Machine Learning Researcher

Denton, TX, USA
October 2022 - December 2023

Responsibilities:

- Performed Data Profiling to learn about behavior with various features such as traffic pattern, location, time, Date and Time etc.
- Created and connected SQL engine through C# to connect database, developed API libraries and business logic using C#, XML and Python
- Exploring DAG's, their dependencies and logs using Air Flow pipelines for automation
- Performed data cleaning and feature selection using MLlib package in PySpark and working with deep learning frameworks such as Tensorflow, PyTorch etc.
- Developed Spark/Scala, Python, R for regular expression (regex) project in the Hadoop/Hive environment with Linux/Windows for big data resources. Used clustering technique K-Means to identify outliers and to classify unlabeled data.
- Worked on supervised learning tasks (e.g., regression, classification) to predict customer lifetime value and identify key factors influencing customer retention.
- Utilized Natural Language Processing (NLP) techniques to analyze customer feedback and extract sentiment insights, improving customer service processes.
- Cleaned, processed, and visualized complex datasets (both structured and unstructured) using Python, SQL, and Excel, helping the team make data-driven decisions.
- Developed dashboards in Tableau/Power BI to visualize key performance metrics for senior leadership, driving informed decision-making.
- Developed a real-time data pipeline using Dataproc (Apache Spark) and Vertex AI, enabling low-latency model inference, reducing processing time from minutes to seconds.
- Built an MLOps framework using Vertex AI Pipelines, automating model retraining based on data drift and performance metrics, ensuring model accuracy remains above 90%.
- Evaluated models using Cross Validation, Log loss function, ROC curves and used AUC for feature selection and elastic technologies like Elastic Search, Kibana etc
- Categorized comments into positive and negative clusters from different social networking sites using Sentiment Analysis and Text Analytics
- Tracking operations using sensors until certain criteria is met using AirFlow technology.
- Responsible for different Data mapping activities from Source systems to Teradata using utilities like TPump, FEXP, MLOAD, BTEQ, FLOAD etc
- Developed and deployed MLflow Models with version control, enabling seamless model serving and rollback capabilities in production environments.
- Analyze traffic patterns by calculating autocorrelation with different time lags.
- Ensured that the model has low False Positive Rate and Text classification and sentiment analysis for unstructured and semi-structured data.
- Addressed overfitting by implementing of the algorithm regularization methods like L2 and L1.
- Used Principal Component Analysis in feature engineering to analyze high dimensional data.
- Performed Multinomial Logistic Regression, Random Forest, Decision Tree, SVM to classify package is going to deliver on time for the new route.
- Performed data analysis by using Hive to retrieve the data from Hadoop cluster, Sql to retrieve data from Oracle database and used ETL for data transformation.
- Used MLlib, Spark's Machine learning library to build and evaluate different models.
- Implemented rule-based expertise system from the results of exploratory analysis and information gathered from the people from different departments.
- Performed Data Cleaning, features scaling, features engineering using pandas and numpy packages in python and build models using SAP Predictive Analytics.
- Developed MapReduce pipeline for feature extraction using Hive and Pig.
- Created Data Quality Scripts using SQL and Hive to validate successful data load and quality of the data. Created various types of data visualizations using Python and Tableau /Spotfire.
- Communicated the results with operations team for taking best decisions.

Client: Tata Consultancy Services
Role: Data Scientist

Karnataka, India
September 2019 - August 2022

Responsibilities:

- Provided Configuration Management and Build support for more than 5 different applications, built and deployed to the production and lower environments.
- Implemented public segmentation using unsupervised machine learning algorithms by implementing k-means algorithm using Pyspark.

- Engineered scalable data pipelines using PySpark and SQL, optimizing ETL workflows for high-volume data processing and reducing latency by 30%.
- Integrated XGBoost and Random Forest models into scalable ML pipelines on GCP Vertex AI, ensuring seamless deployment and real-time inference.
- Analyzed feature importance from Random Forest and XGBoost models to provide actionable business insights, influencing data-driven decision-making.
- Implemented model monitoring and drift detection strategies, retraining XGBoost models periodically to maintain predictive accuracy over time.
- Designed and implemented advanced feature engineering techniques, leveraging domain-specific transformations and statistical methods to enhance predictive model accuracy.
- Fine-tuned BERT and RoBERTa on domain-specific datasets using Databricks MLFlow, optimizing text classification and named entity recognition (NER) models, improving accuracy by 30%.
- Designed ETL pipelines in Azure Data Factory (ADF) to ingest, clean, and transform large text datasets in ADLS Gen2, preparing data for LLM training and fine-tuning in Databricks.
- Built an Azure Synapse-based analytical framework, integrating structured and unstructured data from ADLS Gen2, enabling efficient feature extraction for NLP models, reducing query latency by 25%.
- Deployed fine-tuned LLMs (BERT, RoBERTa) as REST APIs on Azure ML & Databricks, integrating with Azure Functions for real-time sentiment analysis and document classification.
- Developed an MLOps pipeline using Azure DevOps, ADF, and Databricks, automating model retraining when new data arrives in ADLS Gen2, ensuring model freshness and maintaining 90%+ accuracy.
- Processed terabytes of text data using Apache Spark on Databricks, leveraging Synapse for feature engineering, improving NLP model training efficiency by 40%.
- Developed and deployed time-series forecasting models using ARIMA, LSTM, and Prophet, improving demand prediction accuracy by 20%.
- Engineered lag-based and rolling window features to enhance predictive analytics, optimizing model performance for sales and inventory forecasting.
- Developed ensemble forecasting models, combining statistical and ML-based approaches (ARIMA + XGBoost) to enhance robustness and reliability in predictions.
- Using Air Flow to keep track of job statuses in repositories like My SQL and Postgre databases.
- Explored and Extracted data from source XML in HDFS, used ETL for preparing data for exploratory analysis using data munging.
- Responsible for different Data mapping activities from Source systems to Teradata, Text mining and building models using topic analysis, sentiment analysis for both semi-structured and unstructured data.
- Handled importing data from various data sources, performed transformations using Hive, Map Reduce, and loaded data into HDFS
- Used R and python for Exploratory Data Analysis, A/B testing, HQL, VQL, Data Lake, AWS Redshift, oozie, py Spark, Anova test and Hypothesis test to compare and identify the effectiveness of Creative Campaigns.
- Computing A/B testing frameworks, clickstream and time spent databases using Airflow.
- Created clusters to Control and test groups and conducted group campaigns using Text Analytics.
- Created positive and negative clusters from merchant's transaction using Sentiment Analysis to test the authenticity of transactions and resolve any chargebacks.
- Analyzed and calculated the lifetime cost of everyone in the welfare system using 20 years of historical data.
- Developed LINUX Shell scripts by using NZSQL/NZLOAD utilities to load data from flat files to Netezza database.
- Developed triggers, stored procedures, functions and packages using cursors and ref cursor concepts associated with the project using PL/SQL.
- Created various types of data visualizations using R, C#, python and Tableau/Spotfire also connected Pipeline Pilot with Spotfire to create more interactive business driven layouts.
- Used Python, R, SQL to create Statistical algorithms involving Multivariate Regression, Linear Regression, Logistic Regression, PCA, Random Forest models, Decision trees, Support Vector Machine for estimating the risks of welfare dependency.
- Identified and targeted welfare high-risk groups with Machine learning/deep learning algorithms.
- Conducted campaigns and run real-time trials to determine what works fast and track the impact of different initiatives.
- Developed Tableau visualizations and dashboards using Tableau Desktop.
- Used Graphical Entity-Relationship Diagramming to create new database design via easy to use, graphical interface.
- Perform analyses such as regression analysis, logistic regression, discriminant analysis, cluster analysis using SAS programming.
- Developed Bash and PowerShell scripts to automate ETL workflows, enabling seamless data ingestion from multiple sources into cloud storage and databases, reducing manual effort by 50%.
- Integrated Docker with CI/CD pipelines (GitHub Actions, Jenkins, Azure DevOps) to automate model testing, packaging, and deployment, reducing release cycles by 30%.
- Automated ML pipelines using MLflow, orchestrating model training, validation, and deployment with CI/CD integration on GCP Vertex A
- Implemented MLflow Tracking to log model parameters, metrics, and artifacts, ensuring reproducibility and streamlined experiment management.